IEOR 4404: Simulation

Lecture 15: Conditional Monte Carlo

Conditional Monte Carlo

- Suppose we want to estimate $E[g(X)]$, or simply $E[Y]$ where we denote $Y = g(X)$
- Suppose we know $E[Y|Z]$ and can simulate Z
- We output $E[Y|Z]$ as one simulation run output
- With n simulation runs, we output

$$\frac{1}{n}\sum_{i=1}^{n} h(Z_i)$$

where $h(Z) = E[Y|Z]$, and Z_i is the i-th replication

Conditional Monte Carlo

This estimator is called the conditional Monte Carlo estimator

Conditional Monte Carlo is unbiased by the law of total expectation:

$$E\big[E[Y|Z]\big] = E[Y]$$

Each conditional MC outputs:

$$h(Z) = E[Y|Z]$$

$$E[h(Z)] = E[E[Y|Z]] = E[Y]$$

Conditional Monte Carlo

Conditional Monte Carlo has variance no more than naïve MC:

$$Var(E[Y|Z]) \leq Var(Y)$$

because of the law of total variance

$$Var(Y) = Var(E[Y|Z]) + E[Var(Y|Z)] \geq Var(E[Y|Z])$$

Conditional Monte Carlo

- Conditional Monte Carlo has most variance reduction when Y and Z are least dependent

in complete
opposite to
control
variate

- In particular, it has zero variance when Z is independent of Y

- How does it contrast with control variate, and why?

<u>Note:</u>

- Conditional MC is always at least as good as naive MC.

- Control variate is more powerful when Y and Z are more correlated, while conditional MC is more powerful when Y and Z are less dependent.
This discrepancy relates to what knowledge we use for variance reduction.

- when Y and Z are independent.

$$\mathrm{var}(E[Y|Z]) = \mathrm{var}(E[Y]) = 0$$

conditional MC has variance 0.

- when Y and Z are fully dependent, i.e.

$$Y = Z$$

$$\mathrm{var}(E[Y|Z]) = \mathrm{var}(Y)$$

conditional MC has no variance reduction compared to naive MC.

Conditional Monte Carlo

- Conditional Monte Carlo is known to provide significant variance reduction for some heavy-tailed rare-event estimation problems
- Stratification uses the other term in the law of total variance

Example

Estimate $P(X_1 + X_2 > 10)$ where $X_1, X_2 \sim Exp(1)$, by using conditional Monte Carlo

Repeat the above for general distributions

naive MC:

- Generate $(X_1^{(i)}, X_2^{(i)})$, $i = 1, \cdots, n$

- Output

$$\frac{1}{n} \sum_{i=1}^{n} I(X_1^{(i)} + X_2^{(i)} > 10)$$

<u>Conditional MC:</u>

$$P(X_1 + X_2 > 10) = E\left[I(X_1 + X_2 > 10) \right]$$

Consider

$$h(X_1) = E\left[I(X_1 + X_2 > 10) \mid X_1 \right]$$

$$= P(X_1 + X_2 > 10 \mid X_1)$$

$$= P(X_2 > 10 - X_1 \mid X_1)$$

$$= e^{-(10 - X_1)}$$

- Generate $X_1^{(i)}$, $i = 1, \ldots, n$

- Output

$$\frac{1}{n} \sum_{i=1}^{n} e^{-(10 - X_1^{(i)})}$$

This estimator has a variance at most that of naive MC.